IMMUNEBYTES

Security Assessment

CROSS THE AGES

28 July 2025

CTA Token
v1.0

Introduction

Name	CrossTheAges
Website	https://www.crosstheages.com/
Platform	L1
Network	Solana
Language	Rust
Initial Commit	NA
Final Commit	NA
Timeline	25th July 2025 - 28th July 2025

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Overview

This report presents a comprehensive review of the deployment integrity, configuration parameters, and security posture of the Cross The Ages (CTA) token ecosystem deployed across Ethereum and Solana blockchains. The setup leverages Wormhole's Native Token Transfers (NTT) in a hub-and-spoke model, where the Ethereum-based CTA token acts as the canonical asset and the Solana counterpart is dynamically minted through verified Wormhole attestations.

This engagement is not intended to serve as a traditional audit that focuses on a code review. Instead, it delivers an in-depth evaluation of the deployment and configuration process, authority delegation, and associated DevOps practices with a focus on production-readiness, key management, and overall system resilience for a production ready build.

System Overview

The CTA token ecosystem is composed of the following smart contract components:

Layer	Component	Chain	Address
Native Token	ERC20: CTA	Ethereum	0x90685e300A4c4532EFCeFE91202DfE1Dfd572F47
Bridged Token	Token2022: CTA	Solana	CTAyri1T4d1QRfnmtSRpEsPS5tyH6shA1WTskhybmrMT
NTT Manager	Wormhole Manager	Ethereum	0xFB9Cbb83FF576F0530f1BE277dDbD4064bd2717b
NTT Transceiver	Wormhole Transceiver	Ethereum	0xBF53FD055221C354c4c11bD758dcD5dF2822Cd1C
NTT Manager	Wormhole Manager	Solana	CTAvfy818DpTBb4augQr9GSVA1irQfjYdxpUpGfkxaBR
NTT Transceiver	Wormhole Transceiver	Solana	FtH9FXc3XBrN2cUEHXJgBaxk4p9ECXbFeiRR6g8x5zPf
Deployment Wallet	Initial Deployer	Solana	CTAw1SUjArtGuqSZUniipHfsnQayy1m1mXQczi9Z98DW

Ethereum Token Verification

The Ethereum-based CTA token contract has been previously audited by ImmuneBytes. The following observations reaffirm its correct and immutable configuration ensuring a robust deployment setup:

- **Total Supply:** 500,000,000 CTA minted at deployment.
- **Standards Compliance:** Follows OpenZeppelin ERC20 implementation (Solidity 0.8.24).
- **Code Review:** No exposed mint/burn functions; no backdoor administrative controls.
- **Source Verification:** Code is published and verified on Etherscan.

This implementation aligns with best practices for canonical tokens in cross-chain ecosystems.

Solana Token Configuration

The Solana-side Token2022 asset (CTA Token) was instantiated per Wormhole's NTT bridging specification. Our review confirms:

- **Decimals:** 9, consistent with Token2022 conventions.
- **Mint Authority:** Correctly delegated to the Wormhole NTT Manager contract (**CTAvfy818DpTBb4augQr9GSVA1irQfjYdxpUpGfkxaBR**).
- **Freeze Authority:** Freeze Authority has been revoked by the token owner, so no one can freeze the token.
- **Token Authority**: Currently delegated to the deployer address (**CTAw1SUjArtGuqSZUniipHfsnQayy1m1mXQczi9Z98DW**).
- **Deployment Process:** The deploying wallet (**CTAw1SUjArtGuqSZUniipHfsnQayy1m1mXQczi9Z98DW**) followed Wormhole's official initialization guidelines.

These parameters ensure that the token cannot be minted arbitrarily, and that minting flows are securely bound to VAA validations from the Ethereum side.

Wormhole Bridge Behavior

The bridge architecture adopts a hub-and-spoke configuration with Ethereum as the canonical token origin. All bridged token mints on Solana occur via validated Wormhole VAAs.

1. **Ethereum:** CTA is locked by the TokenBridge contract.
2. **Wormhole:** A VAA is emitted and verified by the Solana-side transceiver.
3. **Solana:** CTA SPL tokens are minted to the user via the NTT Manager.

This flow complies with Wormhole's NTT protocol and demonstrates correct cross-chain supply anchoring.

Key Management & Role Delegation

Verified:

- **Mint/Freeze Authority** is securely held by the Wormhole NTT Manager.
- **Deployer Wallet** is not retaining sensitive authorities post-deployment.

Recommendations:

- Convert the Solana deployer account (**CTAw1SUjArtGuqSZUniipHfsnQayy1m1mXQczi9Z98DW**) to a **multisig PDA** to prevent unilateral control.
- Formalize access control policies for any future upgrades or redeployments.

DevOps, Security & CI/CD Best Practices

Given the operational sensitivity of bridging infrastructure, we recommend the following best practices:

Secrets & Authority Management

- All authority keys (signers) should be stored in HSMs or cloud KMS systems.
- Rotate mint and deployer-related keys through secure governance (e.g., multisig).

CI/CD & Deployment Hygiene

- Maintain reproducible builds and deployment logs for traceability.
- Enforce a release validation checklist covering every component from source to runtime.

Concluding Remarks

The current deployment and configuration of the CTA token and its bridging infrastructure appear secure, standards-compliant, and aligned with Wormhole's operational guidelines. Key authorities have been appropriately delegated, and the cross-chain minting logic is trust-minimized.

Minor enhancements, particularly around the governance of the deployment wallet and formalized key control, will further reinforce the system's robustness and production readiness.

This report is provided as a review of deployment integrity, smart contract configuration, and infrastructure-level security assumptions—not a code audit.

Disclaimer

ImmuneBytes' security audit services are designed to help identify and mitigate potential security risks in smart contracts and related systems. However, it is essential to recognize that no security audit can guarantee absolute protection against all possible security threats. The audit findings and recommendations are based on the information provided during the assessment.

Furthermore, the security landscape is dynamic and ever-evolving, with new vulnerabilities and threats emerging regularly. As a result, it is strongly advised to conduct multiple audits and establish robust bug bounty programs as part of a comprehensive and continuous security strategy. A single audit alone cannot protect against all potential risks, making ongoing vigilance and proactive risk management essential.

To ensure maximum security, the project must implement the recommendations provided in the audit report and regularly monitor its smart contracts for vulnerabilities. It is imperative to adopt appropriate risk management strategies and remain proactive in addressing potential threats as they arise.

Please note that auditors cannot be held liable for any security breaches, losses, or damages that may occur after the audit has been completed despite using audit services. The responsibility for implementing the recommendations and maintaining the ongoing security of the systems rests solely with the project.